

Autonomous Financial Advisor

Team Blueprint v2 — With Conversational Voice Agent

Prof. Kadali • MIS 02.303 • Spring 2026 | **Due: Tuesday, May 5, 2026 by midnight**

The risk we are choosing to take

Voice adds a fifth workstream. If voice breaks in rehearsal on 05-04, we demo with a keyboard fallback and mention voice in the Future Directions slide. Voice cannot be allowed to eat time that should have gone to SHAP, the bias audit, or the model card. The Voice/UX Lead's job is to deliver voice AND a keyboard fallback, in that order.

1. The 60-Second Read

We are building a working app that profiles an investor by voice, recommends ETFs, explains the recommendation back to them in spoken plain English, and asks a human to approve before anything is finalized. The app has four AI layers — ML → Generative AI → Agentic → Responsible AI — wrapped in a conversational voice UI. That stack mirrors the course arc (Ch. 1–6), and the voice layer directly responds to Dr. Kadali's stated preference from our last assignment.

Five workstreams, four owners, and one agentic agent. The 04-21 class session is a working session we use for live questions. The build splits cleanly so nobody blocks anybody.

What we owe on May 5:

- Working end-to-end demo (voice profile input → risk tier → ETFs → spoken LLM explanation → Accept/Reject)
- Trained ML classifier with SHAP explainability
- Model card document
- Bias audit with at least one demographic slice
- 10-minute presentation with live demo and Future Directions slide
- Clean code repo with README and setup instructions

2. How the Five Layers Fit Together

Walk this in order with the group. If everyone understands the flow of one user request, the division of labor makes sense on its own.

The user's journey, end to end:

1. The user speaks their profile out loud: "I'm 28, make about 70k, save around 10%, have some student loans, and I'm investing for retirement in maybe 30 years."

2. Voice Layer (speech-to-text) transcribes the audio and a small parsing step pulls out the structured fields: age, income, savings rate, DTI, horizon, experience.
3. Layer 1 (ML) runs inference on those fields. The classifier predicts Low / Medium / High. SHAP explains which features pushed the tier.
4. Layer 3 (Agentic) takes the tier and calls yfinance to build a small ETF shortlist appropriate for that risk level.
5. Layer 2 (Generative AI — Claude) receives the tier, SHAP factors, and ETF list. It writes a 2–3 sentence plain-English explanation.
6. Voice Layer (text-to-speech) speaks the explanation back to the user.
7. Layer 4 (Responsible AI) wraps all of it: SHAP plot visible on screen, model card on file, bias audit documented, mandatory Accept/Reject button (clickable AND voice-acceptable: "accept" / "reject") before anything is logged.

The order ML → Agentic → GenAI → Voice-out matters. The agent needs the tier before selecting ETFs. The LLM needs both tier and ETF list before it can explain anything worth hearing. Voice-out happens last so the user hears a complete thought, not a stream of half-built reasoning.

Data flow at a glance

Layer	Input	Process	Output
Voice (in)	Raw audio	Speech-to-text + field extraction	Structured profile dict
1. ML	Profile dict	Pre-trained classifier + SHAP	Risk tier + feature attributions
3. Agentic	Risk tier	yfinance fetch + tier→ETF mapping	ETF shortlist (ticker, name, price, rationale)
2. GenAI (Claude)	Tier + SHAP + ETFs	Claude API call with guardrailed prompt	2–3 sentence explanation text
Voice (out)	Explanation text	Text-to-speech synthesis	Spoken audio + visible text
4. Resp. AI	All of the above	SHAP display + bias audit + human gate	Accept/Reject decision, logged

3. Tech Stack — Decisions Already Made

If anyone has a strong reason to swap something, raise it at kickoff. After that, no mid-project rewrites.

Component	Choice	Why
Language	Python 3.11+	Every library we need lives here.
UI framework	Streamlit	Fastest path to a demo-able UI. One Python file can render the whole flow.
ML classifier	scikit-learn RandomForest	Handles mixed feature types. Plays nicely with SHAP. Solid baseline without tuning.
Explainability	SHAP (TreeExplainer)	Fast on tree models. Produces the waterfall plot the spec asks for.
LLM (Layer 2)	Claude (Anthropic API)	Layer 2 is user-facing prose. Claude writes warmer, clearer natural language for a non-technical reader. Defensible choice — the spec explicitly leaves this to us.
Speech-to-text	Browser Web Speech API (primary), OpenAI Whisper (fallback)	Web Speech API is built into Chrome/Edge, free, no install. Whisper as local fallback if browser fails.
Text-to-speech	Browser SpeechSynthesis API (primary), ElevenLabs (optional upgrade)	Browser TTS is free and sounds acceptable. ElevenLabs only if time permits — its voice is noticeably better but costs tokens.
Market data	yfinance	Free, no auth, one pip install. Named explicitly in the spec.
Bias audit	pandas + matplotlib	No need for a fairness framework for one slice. Keep it readable.
Version control	GitHub (private) + branch protection on main	One branch per layer. PRs with one approval required to merge to main.

4. Role Assignments

Five workstreams, five owners. If the team is four people, one person double-hats (suggested pairing: Layer 4 + Voice, since the Responsible AI lead's work is front-loaded and Voice is back-loaded). Fill in names together at kickoff.

Role	Owner	Core responsibility
Layer 1 — ML Lead	[name]	Train the risk classifier on financial_risk_profiles.csv Produce SHAP explainer + visualization Export trained model (pickle/joblib) for the app to load Write metrics section of the model card
Layer 2 — GenAI Lead (Claude)	[name]	Design the Claude prompt that takes tier + SHAP + ETFs and returns an explanation Implement the Anthropic API call with API key in env var Enforce guardrails: hedged language, no return figures, 'not financial advice' disclaimer Test for hallucinated tickers (only fund names in the passed ETF list should appear)
Layer 3 — Agentic Lead	[name]	Build tier→ETF mapping (e.g., Low = BND/SCHD, Medium = VTI/VEA, High = QQQ/IWM) Wrap yfinance calls in a retry-safe function with cached fallback for demo day Return a structured payload (list of dicts) Document: recommendation-only, no execution
Layer 4 — Resp. AI Lead / Project Lead	Alex	Run the bias audit on cfpb_financial_wellbeing.csv Author the model card Build the Accept/Reject UI (clickable + voice-acceptable) Own the integration merge on 04-30 and rehearsal on 05-04 Hold the schedule; unblock teammates; submit to Canvas
Voice / UX Lead (new)	[Agentic Agent]	Build the voice-in flow: browser Web Speech API → transcript → field parser Build the voice-out flow: explanation text → SpeechSynthesis Handle mic permissions, empty input, and unrecognized speech gracefully Build the keyboard fallback UI — this is required, not optional Test the full voice loop end-to-end on 04-30 and 05-04

On the bias audit: our training data has no gender, race, or education columns — on purpose. The audit comes from a proxy: bin our training ages into CFPB age groups (under 30 / 30–54 / 55–69 / 70+) and compare the ML model's tier distribution by age band. Document honestly that this is a proxy, not a true demographic fairness test. That honesty is what earns points on Responsible AI — not a polished chart that overclaims what it shows.

5. Timeline — 17 Days to Submission

Two class sessions left (04-21 working session, 04-28 project queries) and one finals week submission.
Two out-of-class meetings per week.

Date	Type	Milestone	Who does what
Sat 04-19	Async	Kickoff prep	Everyone reads this blueprint. Exchange GitHub handles. Confirm Anthropic API access for Layer 2.
Sun 04-20	Meeting 1 (1 hr)	Kickoff	Walk architecture. Lock roles including Voice. Create repo + five branches. Enable branch protection on main. Agree on function signatures (\$6).
Tue 04-21	Class	In-class working session	Bring specific questions to Dr. Kadali, including whether voice satisfies her 'conversational agent' suggestion.
Sat 04-25	Meeting 2 (1 hr)	Layer demos (round 1)	Each owner demos their layer IN ISOLATION. Not integrated yet. Voice demoed standalone with fake inputs.
Tue 04-28	Class	Project queries with professor	Show what we have. Ask about grading weight for the voice stretch vs. the four required layers.
Wed 04-29	Async	Integration begins	Merge to main in order: ML, Agentic, GenAI, then UI+Voice wrapper.
Thu 04-30	Meeting 3 (1.5 hr)	End-to-end voice demo	Full flow runs with voice on one machine. Log every bug. Voice Lead confirms keyboard fallback also works.
Fri 05-01	Async	Bias audit + model card complete	Resp. AI Lead finishes both. Everyone reviews the model card for accuracy about their layer.
Sun 05-03	Meeting 4 (2 hr)	Presentation draft	Build the 10-min deck including Future Directions slide. Assign speaking slots. Practice once end-to-end.
Mon 05-04	Meeting 5 (1 hr)	Dress rehearsal	Time the demo. Run it on the presentation machine. If voice fails here, switch to keyboard. Lock everything.
Tue 05-05	Submit	DUE BY MIDNIGHT	Alex submits to Canvas by 6 PM. No pushes to the repo after that.

6. Agreed Function Signatures

This is the contract that lets parallel work happen. Freeze these at kickoff.

Voice Layer (in)

```
transcribe_profile(audio) -> dict
```

- Input: audio from browser microphone
- Output: {age, annual_income_usd, savings_rate_pct, debt_to_income_ratio, investment_horizon_years, investment_experience}
- If parsing fails, return partial dict + list of missing fields so the UI can re-prompt

Layer 1 — ML

```
predict_risk_tier(profile: dict) -> dict
```

- Output: {tier: 'Low'|'Medium'|'High', confidence: float, shap_values: dict, top_factors: list}

Layer 3 — Agentic

```
get_etf_recommendations(tier: str) -> list[dict]
```

- Output: list of {ticker, name, current_price, category, rationale} — 3 to 5 funds

Layer 2 — GenAI (Claude)

```
generate_explanation(profile, tier, top_factors, etfs) -> str
```

- Output: 2–3 sentence string, hedged, with 'not financial advice' disclaimer

Voice Layer (out)

```
speak(text: str) -> None
```

- Speaks the text via browser SpeechSynthesis while displaying it on screen

Layer 4 — Responsible AI

```
log_decision(profile, tier, etfs, explanation, user_decision)
```

- user_decision can come from a button click OR from a spoken 'accept'/'reject'
- Writes a row to decisions.csv; no decision is logged until user_decision is set

7. The 10-Minute Presentation

Ten minutes, five speakers (one per workstream), ending with Future Directions and a live demo. Every speaker gets ~90 seconds.

Time	Section	Content
0:00–1:00	Opening + Problem	The problem retail investors face. What we built in one sentence. Preview the four layers + voice.
1:00–2:30	Layer 1 — ML + SHAP	Dataset, model choice, accuracy, SHAP plot. One sentence on why RandomForest.
2:30–3:30	Layer 3 — Agentic	How the agent turns a tier into an ETF list. yfinance. Why no live trading.
3:30–4:30	Layer 2 — GenAI (Claude)	Why Claude over GPT for user-facing prose. Prompt design. Guardrails (hallucinated tickers, disclaimer).
4:30–6:00	Layer 4 — Responsible AI	Model card, bias audit methodology and honest caveats, human-in-the-loop gate.
6:00–8:30	Live Demo (voice)	Voice profile entry → tier + SHAP → ETFs → spoken explanation → voice Accept. If voice fails, keyboard demo, no explanation or apology — just move.
8:30–9:30	Future Directions	One slide. Where this could go: CPA study assistant, portfolio tracking, accounting automation. Frame as 'we see the product vision, we scoped the class version deliberately.'
9:30–10:00	Wrap + Q&A cue	Thanks Dr. Kadali. Invite questions.

8. What Could Go Wrong

Risk	Signal	Mitigation
Voice breaks on demo day	Mic denied, browser incompatible, TTS silent	Keyboard fallback is required and tested on 04-30 and 05-04. Demo with keyboard if voice breaks; don't mention it.
Claude hallucinates a ticker	Explanation references a fund not in the agent's list	Pass the ETF list in the prompt as the ONLY allowed universe. Post-check flags unknown tickers.
yfinance errors on demo day	Rate limit, outage, weird market day	Cache a last-known-good response to JSON. Fall back to cache silently.
Integration fails the night before	Layers work alone but not together	04-30 is the integration checkpoint for exactly this reason. 5 days of buffer. If broken on 05-04, we demo the isolated layers and narrate what's missing.
Voice cannibalizes the core four layers	SHAP plot half-built, model card unfinished	Voice Lead stops work on voice if the four required layers aren't green by 04-25. Core scope protects the grade.
Teammate disappears	No response 48+ hrs, missed meeting without notice	Redistribute their layer. Document in peer evaluation honestly — syllabus allows grade adjustment.

9. GitHub Setup (Do This Before Kickoff)

Alex (Project Lead) does this before Sun 04-20 so the group walks into kickoff with a live repo.

8. Create a private repo on GitHub. Name it something like `autonomous-financial-advisor`.
9. Initialize with a README, a Python .gitignore, and MIT license.
10. Settings → Collaborators → add each teammate by GitHub username. They get email invites.
11. Settings → Branches → Add rule for `main`: require pull request, require 1 approval, require branches up to date before merging.
12. Create five branches off main: `layer1-ml`, `layer2-genai`, `layer3-agentic`, `layer4-respai`, `voice-ux`.
13. Enable a Projects board: columns Backlog / In Progress / In Review / Done. Seed it with the deliverables from §10.
14. Set up a Discord or Slack with a GitHub webhook so every push and PR posts automatically. This is the single best investment in visibility — teammates see each other's work within seconds.
15. Add a CONTRIBUTING.md: branch naming, commit message format, PR template, review expectations.

10. Deliverables Checklist

Deliverable	Definition of Done
Working demo	Streamlit app runs with one command. Voice flow OR keyboard flow works end-to-end in under 45 seconds.
ML model	Trained .pkl in repo. Training notebook shows metrics + confusion matrix. SHAP plot renders in the app.
Model card	One to two pages. Training data, features, model type, metrics, intended use, KNOWN limitations, out-of-scope uses, bias audit findings. LLM choice (Claude) and voice layer documented.
Bias audit	Age-band slice. Chart + 2-paragraph finding. Honest about what the proxy can and cannot tell us.
Presentation	10 minutes, 5 speakers, ends with Future Directions slide and live demo. Slides in repo.
Code repo	README setup steps, how to run with and without voice, where each layer lives, env vars for API keys (Anthropic). No secrets committed.

11. Kickoff Meeting Agenda (Sun 04-20)

16. (5 min) Confirm everyone read this blueprint.
17. (5 min) Intensity framing: Alex opens by saying this is a portfolio-grade project. Teammates who want to use it the same way will get visible credit; those who don't, still contribute and still get graded fairly. No pressure, just transparency.
18. (10 min) Walk the architecture diagram (§2). Questions only — no stack debate.
19. (10 min) Fill in names for five roles (§4). Decide if Voice is its own person or a double-hat.
20. (10 min) Confirm tech stack (§3). Anyone without Python, Anthropic API access, or a working mic raises it now.
21. (10 min) Review the function signatures (§6). Frozen after this meeting.
22. (5 min) Walk GitHub setup (§9): everyone confirms they can push to their branch.
23. (5 min) Lock next meeting times (04-25, 04-30, 05-03, 05-04). Put in calendars before leaving.

A Note From the Project Lead

A few things before we start building:

- This is the project I'm treating as portfolio-grade. A working AI system with four integrated layers, a voice interface, a bias audit, and a model card is something any of us can put on a resume or talk about in an interview. If you want to use it the same way, I'll make sure your contribution is visible. If you don't, the grade is still fair — just do your layer well.
- Responsible AI is where student projects usually underperform. Teams build a cool demo and bolt on a one-paragraph 'ethics' section. We are not doing that. The bias audit and model card are started on day one.
- Dr. Kadali's feedback from last time asked for more creativity, specifically a conversational agent with voice. That's why voice is in scope. It's also why I chose Claude for Layer 2 — she has seen generic GPT output a hundred times.
- The Future Directions slide is not fluff. It tells Dr. Kadali and anyone else in the room that we see a product, not just an assignment. Thirty seconds, one slide.
- Bring everything unclear to the 04-21 working session. That is what it's for.

— End of Blueprint v2 —